



Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

INTERNSHIP AT DASSAULT RELIANCE AEROSPACE LTD.

An Internship Report Submitted to
DR. VISHWANATH KARAD MIT WORLD PEACE UNIVERSITY

Submitted by.
SAKSHAM A. SANCHETI
1032220551

Under the supervision of
Prof. Dr. Sandip T. Chavan
(University Guide)
And
Mr. Suhas Warade (Company Guide)



Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Department of Mechanical Engineering
Dr Vishwanath Karad MIT World Peace University, Pune
[2025-26]
(Period from 01/07/2025 to 31/12/2025)



Dr. Vishwanath Karad

**MIT WORLD PEACE
UNIVERSITY** | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS



Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Department of Mechanical Engineering
Dr Vishwanath Karad MIT World Peace University, Pune
[2025-26]

(Period from 01/07/2025 to 31/12/2025)

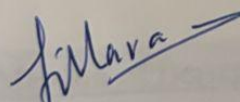
CERTIFICATE

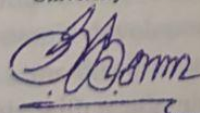
This is to certify that report on

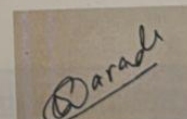
**INTERNSHIP AT DASSAULT RELIANCE
AEROSPACE LTD.**

Submitted by
SAKSHAM A. SANCHETI
(1032220551)

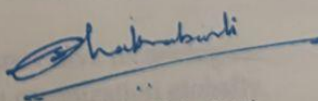
in partial fulfilment of requirements of an Internship at Dassault Reliance Aerospace Ltd., is a bonafide record of the work carried out by him during the period from 01/07/2025 to 31/12/2025. He has worked under the supervision of University Supervisor Prof. Dr. Sandip T. Chavan and Mr. Suhas Warade. He has fulfilled the requirement of the submission of the Internship report for Fourth Year Mechanical Engineering as per the syllabus prescribed by the MIT World Peace University, Pune. The material obtained from other sources has been duly acknowledged in the report.


Prof. Dr. Sandip T. Chavan
University Guide


Prof. Dr. Shivprakash B. Barve
Program Director, Department of Mechanical
Engineering



Mr. Suhas Warade
Company Supervisor


Prof. Dr. Siddharth Chakrabarti
Dean, Engineering and Technology



Vision and Mission

Vision of Faculty of Engineering and Technology

1. To be a globally recognized leader in Engineering Education having a constructive impact on society.

Mission of Faculty of Engineering and Technology

1. To achieve academic excellence through Continuously Updated Education (CUEd).
2. To create environment of technology research and social innovation by establishing trans-disciplinary centres of excellence.
3. To establish strong partnerships with reputed industry and research organizations.
4. To promote ethical and universal value based professional education.

Vision of Department of Mechanical Engineering

To be a centre of academic excellence in Mechanical Engineering for sustainable development.

Mission of Department of Mechanical Engineering

5. To impart comprehensive knowledge in mechanical engineering and foster skills to enhance economic development of the nation.
6. To facilitate trans-disciplinary research leading to innovative technologies.
7. To collaborate with academia, industry, and research organization globally.
8. To nurture students for lifelong learning and ethical professional career.

ACKNOWLEDGEMENT

It gives me great pleasure to present to complete my internship under several hands helped me directly and indirectly. Therefore, it becomes my duty to express my gratitude towards them.

I am very much obliged to my **company supervisor Mr. Suhas Warade** as well as my **University Supervisor Prof. Dr. Sandip T. Chavan**, for helping and giving proper guidance and knowledge. Their timely suggestions made it possible to complete this internship

I will fail in my duty if I will not acknowledge a great sense of gratitude to the program director of Department of Mechanical Engineering **Prof. Dr. Shivprakash B. Barve** and the entire staff members in the Department of Mechanical Engineering for their cooperation.

Saksham A. Sancheti

PRN No-1032220551

Department of Mechanical Engineering,
MIT WPU, Pune

DECLARATION

I the undersigned, declare that the work carried out under the title entitled “Internship at Dassault Reliance Aerospace Ltd.” represents my idea in my/our own words. I have adequately cited and referenced the original sources where other ideas or words have been included. I also declare that I have adhered to all principles of academic honesty and integrity and have not misprinted or fabricated or falsified any ideas/facts/source in my/our submission. I understand that any violation of the above will be cause for disciplinary action by the University and can also evoke penal action from the sources which have thus not been properly cited or for whom proper permission has not been taken when needed.

Saksham A. Sancheti

PRN No-1032220551

Department of Mechanical Engineering,

MIT WPU, Pune

ABSTRACT

During the six-month internship at **Dassault Reliance Aerospace Ltd.**, Mihan–Nagpur, **practical experience** was gained across the spectrum of **aerospace manufacturing**, with a special focus on the **assembly processes** for **Falcon and Rafale aircraft**. Key responsibilities encompassed streamlining production workflows, **optimising shop floor manufacturing activities** and deploying robust **inventory management solutions** to achieve traceability. Advanced fabrication methods, notably **FDM 3D printing**, were utilized to create operational fixtures and components integrated within live manufacturing environments. The internship also provided exposure to essential industry practices, including **Foreign Object Debris (FOD) prevention**, **Safety Management Systems (SMS)**, **Standard Operating Procedures (SOPs)**, **Process Descriptions (PD)**, **Quality Assurance Plan (QAP)**, **Production Organization Exposition (POE)** interpretation of engineering drawings, and Manufacturing Bill of Materials (MBOMs) verification. This experience fostered a deep, practical understanding of **aerostructure production**, process optimization, and quality control aligned with **global aerospace standards**, while significantly developing the technical, analytical, and collaborative skills essential for a successful aerospace engineering career.

INDEX

Sr No.	Contents	Page No.
--------	----------	----------

1.	Company Overview 1.1 Background of the Company 1.2 Products and Services 1.3 Historical Overview of the Company	
2.	Literature Review 2.1 Introduction 2.2 Overview of Aerospace Manufacturing 2.3 Aerostructure Assembly and Integration 2.4 Additive Manufacturing in Aerospace Production 2.5 Standard Operating Procedures (SOPs) and Quality Assurance 2.6 Inventory Management and Traceability Systems 2.7 Key Performance Indicators (KPIs in Aerospace Production) 2.8 Summary	
3.	Methodology/ Approach 3.1 Problem Statement I 3.1.1 Methodology 3.1.2 Discussion 3.1.3 Summary 3.2 Problem Statement II 3.2.1 Methodology 3.2.2 Discussion 3.2.3 Summary 3.3 Problem Statement III	

	3.3.1 Methodology 3.3.2 Discussion 3.3.3 Summary 3.4 Problem Statement IV 3.4.1 Methodology 3.4.2 Discussion 3.4.3 Summary	
4.	Conclusion	
5.	Benefits to the Company	
6.	Evaluation and Feedback by Company Supervisor	

LIST OF FIGURES

Fig No.	Name of figure	Page No.
1	DRAL Logo	1
2	Falcon 2000 Business Jet	2
3	RAFALE Fighter Jet	2
4	Falcon 8X Business Jet	2
5	Assembly of T1 cabin at DRAL	5
6	Personal Picture at DRAL Facility	26
7	Evaluation by Company Guide	27
8	Feedback by Company Guide	28

CHAPTER 1- INTRODUCTION

1.1 Background of the Company:

Dassault Reliance Aerospace Ltd. (DRAL) is a joint venture between Dassault Aviation of France and Reliance Aerostructure Limited, part of the Indian Reliance Group. Established in 2017, DRAL plays a pivotal role in advancing India's aerospace manufacturing capabilities, particularly in aerostructure assembly for Falcon business jets and Rafale fighter aircraft.

Company Overview:

- **Company Name:** Dassault Reliance Aerospace Ltd.
- **Company Location:** SEZ Mihan, Nagpur, India
- **Year Established:** 2017
- **Type:** Joint venture (51% Dassault Aviation 49% Reliance Infrastructure,)
- **Specialization:** Aerostructure Assembly
- **Company Vision:** To be a preferred partner of choice for building business jets and fighter aircraft
- **Company Mission:** To build an aircraft meeting global standards and facilitate aerospace ecosystem to increase shareholders value



Figure 1: DRAL LOGO

1.2 Products and Services:

Dassault Reliance Aerospace Ltd specialises in the manufacturing of major aerostructure assemblies for Business and fighter jets:

1] Falcon 2000 Business jet



2] RAFALE Fighter jet



3] FALCON 8X Business jet



CHAPTER 2 - LITERATURE REVIEW

2.1 Introduction

The aerospace industry demands **precision, reliability, and adherence to global quality standards**. Modern manufacturers like **Dassault Reliance Aerospace Limited (DRAL)** integrate advanced production practices, digital monitoring, and lean documentation to enhance efficiency and traceability. This chapter reviews key concepts relevant to the internship:

- aerostructure assembly
- FDM-based additive manufacturing
- SOP development
- inventory management
- KPI tracking

This literature review aims at providing a theoretical foundation for understanding their role in contemporary aerospace manufacturing system.

2.2 Overview of Aerospace Manufacturing

Aerospace manufacturing focuses on building aircraft parts and systems with extremely **high precision and reliability**. Every component — from small brackets to large fuselage structures — must meet **strict airworthiness and safety requirements**, which makes the industry unique and **heavily regulated**.

Companies such as Dassault Aviation, Airbus, and Boeing follow global standards like **AS9100**, **NADCAP** and various **EASA norms** to maintain consistent quality and ensure that every process is traceable and audit-ready. These standards help eliminate errors, improve safety, and create a structured workflow across suppliers and production teams.

In recent years, **aerospace factories** have steadily moved toward digital and smart manufacturing. Concepts from **Industry 4.0** such as **digital documentation, automated data tracking, and live process monitoring** are becoming part of everyday shop-floor activities. These advancements help engineers and technicians work more efficiently, reduce paperwork, and maintain better traceability.

At its core, aerospace manufacturing blends **precision engineering, strict quality control, and modern automation**. This environment sets the foundation for hands-on industrial learning and provides interns with real-world exposure to advanced production systems and quality-driven processes.

2.3 Aerostructure Assembly and Integration

Aerostructures include major aircraft components such as **wings, fuselage sections, and tail** assemblies. These structures are assembled using precision jigs, fixtures, and alignment tools to ensure accuracy and structural integrity.

The assembly process involves operations like fastening, sealing, fitting, and bonding: all performed under clear Standard Operating Procedures (SOPs) and quality checks. Even a small deviation can affect aircraft performance, which is why repeatability and discipline are highly valued on the shop floor.

Research and industrial best practices highlight the importance of ergonomics, process standardization, and lean principles such as 5S and Kaizen. These systems help technicians work comfortably, reduce waste, and maintain consistent quality.

For students and trainees, aerostructure assembly provides a real-world understanding of how engineering drawings, manufacturing standards, and teamwork translate into a flight-worthy product.



Figure 5: Assembly of Cabin [T1] at DRAL

2.4 Additive Manufacturing in Aerospace Production

Additive Manufacturing (AM) — especially Fused Deposition Modelling (FDM) — plays an increasingly important role in aircraft production support. While critical flight components are mostly metallic and composite-based, polymer 3D printing is widely used for:

- jigs and fixtures
- assembly & inspection aids
- ergonomic tool designs
- prototype parts

The advantage of FDM lies in its **low cost, quick turnaround, and design flexibility**, which allows engineers to iterate quickly and solve shop-floor challenges efficiently.

Organizations like NASA and EASA have shared studies showing how polymer AM can simplify tooling requirements and improve flexibility in manufacturing environments. At facilities such as DRAL, **FDM supports day-to-day production and reduces delays** by offering **custom solutions in-house**.

Technical Background for FDM based 3D printing at DRAL mainly include the concepts of Infill Density, Custom support generation, and accommodation for shrinkage.

One of the most important considerations in FDM printing is **infill density**, which directly affects part strength, weight, and print duration. For lightweight prototype or visual fit-check models, infill levels between **10–30%** are typically sufficient. In contrast, functional shop-floor tooling or load-bearing support

components often use **40–100% infill** to achieve increased rigidity and durability. This ability to tune part properties helps engineers balance strength requirements with material efficiency and printing cost.

Support structures are another vital aspect of production-grade 3D printing. Standard automatic supports work for simpler geometries, but **custom support generation** is often preferred in aerospace environments to reduce post-processing time and protect critical surfaces from damage during removal. Thoughtful support design improves dimensional accuracy, enhances surface finish on overhangs, and reduces rework time.

A common challenge in polymer-based additive manufacturing is **thermal shrinkage and dimensional deviation**, especially in larger components or geometries with sharp transitions. To overcome this, engineers rely on calibrated machine settings, controlled bed and nozzle temperatures, and strategic print orientation to minimize warping. Techniques like **adaptive slicing**, **draft angle modifications**, and **fillet additions** further help maintain geometric stability.

Post-processing and rectification techniques also ensure that printed components meet functional requirements. These include **sanding, deburring, precision trimming, and hole-calibration** using drills or reamers for accurate fits. In certain cases, surface finishing methods such as vapour polishing or epoxy coating are used to enhance wear resistance and surface quality for shop-floor tools.

For Additive manufacturing taking place at DRAL, Fused Deposition Modelling based printer: **Ultimaker Extended 3 Double Nozzle Printer** is used with PLA

0.4mm material spool for component fabrication alongside Nylon 0.4mm water soluble spool for support generation.

Design is made through CATIA V5 software, and the .stl file generated is transferred to Ulimaker Cura Software for Slicing, Parameter setting, and G-Code generation.

2.5 Standard Operating Procedures (SOPs) and Quality Assurance

Standard Operating Procedures (SOPs) form the backbone of aerospace manufacturing. They provide step-by-step instructions for every process, ensuring consistency, safety, and compliance.

In modern aerospace environments, SOPs are increasingly digital — equipped with **visual instructions, revision histories, and approval workflows**. This digital approach minimizes ambiguity and keeps the workforce aligned with current standards and revisions.

Effective SOP drafting supports the **broader Quality Management System (QMS)** and encourages accountability among operators and technicians. For interns, contributing to **SOP development** builds a **strong foundation in structured engineering communication**.

2.6 Inventory Management and Traceability Systems

Traceability is essential in aerospace: every tool, part, and material needs to be **tracked** throughout its **entire lifecycle**. Modern factories use:

- barcode-based systems
- RFID tagging
- ERP-integrated digital inventory platforms
- MBOM Verification

These systems **improve audit readiness**, prevent loss or mix-ups, and help **identify root causes** during quality investigations.

During the internship, the implementation of **Excel-based macro tools for traceability, documentation using high resolution camera and stock monitoring** demonstrated a simple yet powerful approach to **digital tracking**, enabling quick insights into **inventory flow and tool usage**.

Additionally, a SOP was also developed to standardise the procedures for advanced Inventory management systems at Dassault Reliance Aerospace Ltd.

2.7 Key Performance Indicators (KPIs) in Aerospace Production

KPIs help measure **production performance** and **identify improvement** areas.

Common aerospace KPIs include:

- On-Time Delivery (OTD)
- Mean Time Between Failures (MTBF)
- Inventory Turnover

Digitizing KPI tracking through **macro-enabled Excel** dashboards allows managers and engineers to view **live shop-floor performance trends** and make informed decisions quickly. Such tools support lean improvement and build a culture of data-driven operations.

2.8 Summary

This literature review highlights how **aerospace manufacturing** combines **precision engineering** with **strict quality systems**, lean practices, digital tools, and modern additive manufacturing techniques. The **internship** experience at **Dassault Reliance Aerospace Limited** aligned with these principles through:

- aerostructure assembly exposure
- 3D-printed tooling development
- SOP creation and revision
- digital inventory and KPI tracking

Overall, the work contributed to a **deeper understanding** of **structured manufacturing environments** and the evolving role of digital technologies in **aerospace production**.

CHAPTER 3 - METHODOLOGY / APPROACH

3.1 Problem Statement 1

Advanced Design, Development & Implementation of Complex Additively Manufactured Components for Aerospace Drilling Systems

The composite and titanium drilling operations for Falcon aircraft sections required specialized nose modules capable of providing chip evacuation, coolant channelling, air seal integrity, and ergonomic support. Existing units showed limitations in capture efficiency, sealing accuracy, and maintenance requirements. Additionally, geometric complexity prevented conventional machining approaches, and temporary workarounds affected repeatability and operator efficiency.

To meet Dassault Aviation's global production requirements, a new set of **complex AM-based interface structures** with internal cooling channels, controlled airflow paths, and multi-ridge sealing geometry were designed, validated, and deployed. This initiative required **global collaboration with Dassault Aviation (France) and DASI engineering**.

3.1.1 Methodology

Technical Design & Engineering

- Conducted **hole-quality analysis, chip morphology study, coolant flow behaviour review**
- Designed parts using **CATIA V5** with:
 - Multi-axis sealing geometry
 - Internal vacuum & coolant pathways
 - Datum bossing & anti-rotation locking ribs
 - Load-bearing lattice zones with 65–90% variable infill

- Performed **MBOM verification** & interface compliance with drilling units (Cleco/Desoutter)
- Evaluated **GD&T tolerances**, considering composite contact interfaces & vacuum sealing

Additive Manufacturing Workflow

- Material chosen: **CF-PLA / PLA-HT hybrid blend** for rigidity & thermal stability
- Printing strategies:
 - **Hybrid tree + custom supports** for internal cavities
 - Adaptive slicing at 0.18mm layer resolution
 - Shrinkage offsets, build-plate thermal profiling
 - Reinforcement ribs for stress distribution
- Post-processing:
 - Hole reaming, sealing edge polishing
 - Dimensional correction & surface finishing
 - Fitment & airflow validation on shop floor

Global Industrial Collaboration

- Weekly review & validation calls with **Dassault France & DASI**
- Design sign-offs via structured industrial workflow
- Documentation aligned to Dassault Aviation global methodology

On-Floor Deployment

- Directed 5–10 fitters during installation & live-testing
- Conducted **operator training** & iterative refinement
- Validation under production loads and coolant flow cycles

Documentation & Standards

- SOP creation aligned with **Dassault global standards**
- Version control, QR revision tagging, operator safety notes

3.1.2 Discussion

The project integrated design engineering, additive manufacturing, production coordination, and global aerospace communication. Complex internal channels, shrinkage compensation, and sealing interface tolerances highlighted advanced AM application beyond prototyping, demonstrating production-ready additive engineering.

3.1.3 Summary

Delivered a full end-to-end AM solution deployed in real production, improving hole quality, reducing chip contamination, improving ergonomics, and cutting cycle time by ~20–25%. Strengthened expertise in **CATIA V5, AM strategy, MBOM validation, AS9100 workflow, and international technical collaboration.**

3.1 Problem Statement 2:

Pneumatic Conversion for Fuel-Tank Assembly Jig to Enhance Repeatability & Ergonomics

Fuel-tank jig operations required manual mechanical actuation, leading to operator fatigue, force inconsistency, and longer assembly cycles. A pneumatic conversion was necessary to meet industrial safety, productivity, and ergonomic standards while aligning with Dassault Aviation process benchmarks.

3.2.1 Methodology

Requirement Study & Concept Design

- Evaluated existing manual locking mechanism and load conditions
- Conducted ergonomic and cycle-time study on assembly sequence
- Designed a pneumatic actuation solution to replace manual sliders/locks
- Selected **double-acting pneumatic cylinders (Festo brand)** based on force, stroke, and duty-cycle requirements
- Created pneumatic circuit layout including **T-valves, flow regulators, and quick connectors**
- Developed bill of materials (BOM) and ordered all components

Simulation & Functional Testing

- Assembled prototype system for functional validation
- Verified **simultaneous and synchronized actuator motion** to ensure balanced clamping force
- Tested for pressure stability, response time, and system integrity
- Made iterative adjustments to valve configuration and flow controls

Approvals & Documentation

- Prepared technical justification and implementation plan
- Conducted presentation/demo for Industrialisation and Quality teams
- Obtained necessary approvals from production leadership and QA
- Created installation documentation, pneumatic routing sheet, and operator guidelines

Installation & Deployment

- Installed pneumatic system on the actual jig
- Secured air-line routings and fittings with EHS compliance
- Performed final calibration and leak checks
- Deployed system to production with operator briefing

3.2.2 Discussion

The pneumatic upgrade eliminated manual force, improved consistency in clamping, and enhanced assembly ergonomics. Using **dual synchronized double-acting cylinders** ensured uniform load distribution during fuel-tank clamping, improving repeatability. Integration with production systems required cross-functional collaboration, and attention to safety protocols ensured reliable deployment. The project demonstrated hands-on experience in industrial pneumatics, jig modification, vendor interaction, and aerospace-grade implementation workflow.

3.2.3 Summary

The conversion project successfully transformed the fuel-tank jig from manual to automated operation. Key achievements included:

- Improved operator ergonomics & safety

- Higher repeatability & process stability
- Reduced fatigue-related variability
- Successful implementation using industrial-grade pneumatic components (Festo)
- Complete cycle from design → procurement → testing → approval → deployment

This project reinforced competencies in **pneumatic system design, jig integration, shop-floor validation, and aerospace production practices.**

3.3 Problem Statement 3:

End-to-End Digital Traceability & Production Readiness System for T15 Fuselage Assembly — Global Coordination, MBOM Validation & Asset Deployment

The Falcon T15 fuselage assembly line at DRAL required full tool and equipment readiness before commencement of structured production. However, the tool ecosystem (approx. **400+ specialized aerospace assembly assets**) lacked unified traceability, digital documentation, and status verification.

Thus, a fully traceable & production-ready system was required — aligning with **Dassault Aviation global standards and industrial readiness protocols**.

This responsibility was assigned to me as the **primary owner**, working with:

- Dassault Aviation global team (France)
- DASI industrialisation personnel
- DRAL Production, SCM, PPC, and QA
- **Leadership of 5–10 fitters over ~2 months**

3.3.1 Methodology

A. Tool & BOM Engineering

- Retrieved and interpreted **MBOM (Manufacturing BOM)** for T15 line
- Mapped tools vs station requirements (Station-wise deployment matrix)
- Verified mandatory & optional tooling lists

- Coordinated with DA/DASI for tooling clarification & validation

B. Physical Audit & Technical Validation

- Exhaustive inspection of ~400 tools, fixtures & gauges
- Condition check (mechanical integrity, wear, alignment, calibration needs)
- Verification of:
 - Torque tools & certification status
 - Alignment pins, jigs, sealant tools, fastener tools
 - Safety & ergonomic compliance

C. Coding, Tagging & Traceability

- Developed **aerospace-standard tool ID system**
- Linked IDs to MBOM & workstation deployment plan
- Tagging: Metal tags, printed labels & digital ID registry
- QR coding pilot for future ERP integration (SAP-Hyena 4 compatibility)

D. Digital Data System Development

- Created **macro-enabled Excel traceability system** containing:
 - Tool ID, category, location, station mapping
 - Health condition & calibration status
 - Deployment status (Ready/Repair/Pending)
 - Observations & rectification notes
 - Audit compliance checklist
- Designed **filterable dashboards** for leadership visibility

E. Execution & Team Leadership

- Directed **5–10 fitters** for continuous inspection & tagging
- Shop-floor deployment & rack structuring
- Daily reporting to Production & Industrialisation
- Weekly sync with Dassault France team

F. Audit & Handover

- Prepared line readiness dossier for QA & CQO review
- Supported **internal and OEM-level audits**
- Submitted digital master file to Industrialisation & Quality

3.3.2 Discussion

The T15 readiness project operated at the intersection of:

Domain	Contribution
Production engineering	Ensuring tools align with station tasks & MBOM
Industrialisation	Validating the tool deployment for new line setup
Quality & compliance	Audit readiness, traceability, AS9100 discipline
Global coordination	DA France & DASI India interface
Shop-floor leadership	Managing & guiding 5–10 fitters
Digital transformation	Macro dashboards, QR traceability mapping

This was a **line-readiness & traceability program** critical to:

- Launching Falcon fuselage assembly activities
- Passing supplier readiness checks from Dassault Aviation
- Ensuring uninterrupted tool availability
- Strengthening audit & compliance posture

- Improving coordination across Production-SCM-PPC-QA-Industrialisation

The work directly influenced **production uptime, audit success, and global quality confidence**, establishing a template for future line setups at DRAL.

3.3.3 Summary

This project provided end-to-end ownership of a critical industrial responsibility. It required:

- Engineering competency (MBOM, tool interpretation, aerospace tooling)
- Digital skill (macro dashboards, coding system)
- Shop-floor leadership (team handling & deployment)
- Global communication (DA-France & DASI collaboration)
- Quality & compliance discipline (AS9100 traceability & audit readiness)

Outcome:

A fully verified, traceable, and ready-to-deploy tool ecosystem for T15 line, supporting aircraft manufacturing continuity and Dassault quality standards.

This initiative demonstrated the ability to **bridge engineering, operations, and global aerospace processes** — and marked the capstone of the internship.

3.4 Problem Statement 2:

SOP Development & Process Standardization as per Dassault Aviation Global Practices

Process instructions required clarity, alignment with global formats, and improved visual guidance to support operators and ensure AS9100 compliance.

3.4.1 Methodology

- Benchmarked Dassault Aviation document architecture
- Introduced revision control, document number standardization
- Added visual steps, torque tables, tool IDs, safety markers
- Integrated operator feedback & training checkpoints
- Reviewed legacy SOPs, process docs & work instructions

3.4.2 Discussion

Improved communication, reduced ambiguity, and enhanced compliance.
Learned global aerospace documentation philosophy and quality discipline

3.4.3 Summary

Standardized instructions, improved shop-floor knowledge transfer, and established documentation aligned with global aviation standards

CHAPTER 4 - CONCLUSION

The internship at Dassault Reliance Aerospace Limited provided valuable hands-on exposure to aircraft manufacturing operations, global aviation standards, and structured industrial practices. Working on the Falcon program allowed me to contribute to the T15 fuselage assembly line readiness through complete tool traceability setup, MBOM verification, and cross-department coordination, including interaction with Dassault Aviation and DASI teams. Additionally, I gained practical experience in designing and deploying advanced 3D-printed components, converting a manual jig to a pneumatic system, and assisting in SOP development aligned with aerospace quality requirements. This internship strengthened my technical knowledge, problem-solving ability, and teamwork skills, while reinforcing my ambition to pursue a career in the aerospace sector

CHAPTER 5 - BENEFITS TO DASSAULT RELICANCE AEROSPACE LTD.

- Supported T15 line readiness through structured asset verification and deployment
- Reduced search time and improved operational efficiency on the shop floor
- Enhanced ergonomics and repeatability by converting manual systems to pneumatic operation
- Improved machining consistency with advanced additive-manufactured components
- Strengthened documentation quality in line with Dassault global standards
- Facilitated cross-functional coordination and audit preparedness
- Contributed to continuous improvement and production discipline



Personal Picture at DRAL Facility

CHAPTER 6: EVALUATION AND FEEDBACK BY COMPANY GUIDE

Internship Logbook

EVALUATION BY COMPANY SUPERVISOR

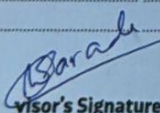
COPYRIGHT OFFICE
NEW DELHI
Reg. No. 194123/2020
Date 26.08.2020

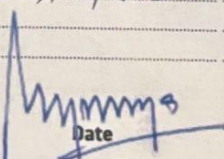
Please assess the internship work on the following 5-Point Scale:
5=Excellent 4=Good 3=Satisfactory 2=Below Average 1=Poor

Sr. No.	Parameters	Excellent (5)	Good (4)	Satisfactory (3)	Below Average (2)	Poor (1)
1	Ability to understand a practical situation & formulate a problem	✓				
2	Ability to collect, collate and analyze appropriate data	✓				
3	Ability to apply theoretical concepts to practical situations	✓				
4	Ability to work independently and demonstrate pro-activeness		✓			
5	Working as part of a Team	✓				
6	Oral Communication	✓				
7	Written Communication	✓				
8	Attendance		✓			
9	Punctuality	✓				
10	Suitability of Office Attire	✓				
11	Behavior	✓				
12	Employability Skills	✓				
13	Flexibility	✓				
14	Organizing own work	✓				
15	Attitude to work	✓				
16	Relationship with Supervisor	✓				
17	Quality of Internship Work	✓				

Additional comments

Overall performance during Internship tenure is excellent from cooling point of view.

Supervisor's Signature: 

Date: 

उप पंजीयन अधिकारी प्रतिनिध्याधिकार
DEPUTY REGISTRAR OF COPYRIGHT

28

OVERALL ASSESSMENT

COPYRIGHT OFFICE
NEW DELHI
No. - L-94123/2020
Date 28.08.2020

The Intern benefits from formal feedback of their strengths, weaknesses and achievements during the internship.

The company supervisor is requested to write it below:

Saksham's has shown noticeable growth during this review period. His approaches his work with Sincerity and willingness to learn, which reflects in the quality of output.

He is dependable when given responsibility and ensure tasks are completed accurately and on time.

* Key reviewed Area-

- supported daily tooling operation by following established procedures and maintaining compliance with Dassault Aviation Standard.

- took initiative in improving work documentation and maintaining traceability, which helps streamline tooling activity.

Overall, Saksham has been a consistent and sincere performer during this period. His willingness to learn and his reliability make him a valuable part of team. With continued exposure and guidance, he is expected to grow further and take on more complex role and responsibility.

ALL THE BEST 😊



Supervisor's Signature

(Signature)

29

(Signature)
Date

उप पंजीयन अधिकारी प्रतिनिध्याधिकार
DEPUTY REGISTRAR OF COPYRIGHT

REFERENCES

- 1] Dassault Aviation (2025): Public Domain url website
- 2] Deloitte. (2021). *Aerospace and Defense Manufacturing Outlook*. Deloitte Insights.
- 3] ISO/AS9100 Rev D. (2016). *Quality Management Systems — Requirements for Aviation, Space and Defense Organizations*.
- 4] NASA. (2022). *Additive Manufacturing Applications in Aerospace Structures and Tooling*.
- 5] NASA Technical Report Series.
- 6] EASA. (2021). *Part-21 & Part-145 Safety and Manufacturing Compliance Framework*. European Union Aviation Safety Agency.
- 7] Ultimaker. (2023). *FDM Additive Manufacturing Guidelines for Engineering Components*.
- 8] Ultimaker Technical Whitepaper.
- Festo AG & Co. KG. (2023). *Pneumatic Actuation System Selection, Control & Safety Manual*. Festo Industrial Automation Standards.

Annexure 1: Internship Request Letter



Ref. No./MITWPU/B.Tech/Mech & RA/Internship/2025-26

Date: 12/02/2025

To

**The Manager,
HR Department,
Dassault Reliance Aerospace Ltd,
3-1A, Sector 9,
Dhirubhai Ambani Aerospace Park (DAAP),
SEZ, MIHAN,
Nagpur,
Maharashtra 440019**

Subject: Internship for Final Year B.Tech. Mechanical Engineering Student: Saksham Sancheti

Respected Sir/Madam,

Maharashtra Academy of Engineering & Educational Research (MAEER) was founded in June 1983 by stalwart and eminent academician Hon. Dr. Vishwanath D. Karad with the collective efforts of many devoted experts having long and outstanding academic experience in the field of professional education. **Maharashtra Institute of Technology, Pune, INDIA** an engineering institute set up by MAEER Pune in 1983 is considered to be a benchmark in engineering education. **MIT Pune is now MIT-World Peace University.** The university focuses on transforming a young, enthusiastic student into a professionally competent engineer capable of accepting the challenges of the industry. Ample opportunities are provided for interaction with the experts from the industry through guest lectures, field visits, vacation training, internships, activities of the student chapters of international professional bodies, sponsorships to technical paper presentation competitions, etc. MIT-WPU Pune is committed to impart a value based universal education to develop 'winning personalities'. At MIT, we focus on excellent infrastructure and faculty for imparting highest quality of holistic education and emphasize on inculcating the right values through discipline and spiritual education.

As part of educational curriculum, the School of Mechanical Engineering, MIT-World Peace University, Pune students have to do internship in various industries like your esteemed organization.

The internship will expose the students to the industrial environment, which cannot be simulated in the classroom and hence creating competent professionals for the industry. The internship will provide possible opportunities to learn, understand and sharpen the real time technical skills required at the job. The students will have exposure to the current technological developments relevant to the subject area of training.



Four Decades of Educational Excellence

The internship period is from **1st July 2025** for period of **6 Months**.

We request you to grant permission for doing internship in your esteemed organization for our student.

A line of confirmation will be highly appreciated.

With warm regards,
Yours sincerely,


Prof. Mahesh Pradhan
Internship Co-ordinator
Department of Mechanical Engineering
MIT WPU


Prof. Dr. Shivprakash B. Barve
Program Director
Department of Mechanical Engineering
MIT WPU

Student information:

Name	Address	Mobile No.	Parent's Mobile No.	Email ID
Saksham Ashok Sancheti	A-1401, Mudra Society Pune Satara Road Pune-411037	8007030658	9422085402	strsaksham@gmail.com

Department Co-ordinator information:

Name	Address	Mobile No.	Email ID
Prof. Mahesh Pradhan	Department of Mechanical Engineering, MIT-WPU	9673740757	mahesh.pradhan@mitwpu.edu.in

Annexure 2: Offer Letter from Company:



Ref : DRAL/TA/2025/002
Date : 25 Feb 2025

To,
Mr. Saksham
C/o. MIT World Peace University, Pune.

Subject: Letter of Internship

Dear Saksham,

With reference to the recommendation letter received from your college and interaction we had with you, we are pleased to inform that you have been selected for Internship in Production Department of Dassault Reliance Aerospace Limited (DRAL) on following terms and conditions.

1. Your internship will start from 01-July-2025. Any change in the date of starting of internship need to be agreed in writing.
2. Your internship location would be DRAL factory in Nagpur till further advise.
3. Period of Internship will be from 01-July-2025 till close of working hours on 31 Dec-2025.
4. You understand and agree that participating in Internship Program is not an offer of employment in the company.
5. During the Internship you will have access for Company's documents and confidential information as part of your internship. You agree that you will keep all the information strictly confidential and will not share to any of your colleague and relatives or any other persons of institutes, also agree to sign confidentiality agreement.
6. You will agree that you will not share any information / updates about company in any of the social media websites. You will be required to sign confidential undertaking upon your joining.
7. You will comply with company's policies, rules & regulations that are in place and may be introduced from time to time.



DASSAULT RELIANCE AEROSPACE LIMITED

Corporate Office & Works: Plot 3-1A, Dhirubhai Ambani Aerospace Park, Sector 9, MIDHAN SEZ, Nagpur - 441 108.
Regd. Office: 502, Plot no. 91/94, Prabhat Colony Santa Cruz (East), Mumbai, Maharashtra 400055
CIN NO U39999MH2017PLC291083; Tel: 0712- 2880 000



As a token of acceptance, you are requested to sign this letter and provide consent for acceptance of terms and conditions mentioned above.

Thanking You,
For Dassault Reliance Aerospace Ltd.


David Raju
Head - Human Resources and Admin.

I accept the terms and conditions mentioned above and I will abide by company's rules & regulations. And I will join the date mentioned in this letter.

Name: SAKSHAM SANCHETI

Signature: 

Date: 9/03/2025

DASSAULT RELIANCE AEROSPACE LIMITED
Corporate Office & Works: Plot 3-1A, Dhairubhai Ambani Aerospace Park, Sector 9, MIDHAN SEZ, Nagpur - 441 108.
Regd. Office: 302, Plot no. 91/94, Prabhakar Colony Santa Cruz (East), Mumbai, Maharashtra 400055
CIN NO U15999MH2017PLC291083